

Machine Learning-Assisted Multiobjective Optimization of Advanced Node Gate-All-Around Transistor for Logic and RF Applications

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Abstract—In this work, using multiobjective optimization (MOO) technique, design optimization of a gate-all-around field-effect transistor (GAAFET) has been performed for improved device logic and RF parameters. By using fast and accurate machine learning (ML) surrogate model, we have emulated the logic and RF performance figures of merit as analytic functions of the design objectives. Datasets required to train and test the ML model are generated using the well-calibrated TCAD setup. The multiobjective optimizers automate and perform extremely fast multispace design optimization. Contrary to MOO, TCAD optimization is tedious and time-intensive. Keeping in view of the International Roadmap of Devices and Systems (IRDS) target, optimal design trade-offs between I_{ON}/I_{OFF} ratio and speed for logic; gain and cut-off frequency (f_T) for RF operation are obtained for a sub-2 nm node GAAFET. Circuit simulation is performed to further validate the design optimization methodology. Moreover, it has been demonstrated that an efficient and faster trade-off between complex nonlinear design parameters can be automated by leveraging the ML-coupled MOO framework for any advanced-node FET.

Index Terms—Artificial neural network (ANN), gate-all-around field-effect transistor (GAAFET), machine learning (ML), multiobjective optimization (MOO), nondominated sorting genetic algorithm (NSGA-III), PKID, strength Pareto evolutionary algorithm (SPEA2).

I. INTRODUCTION

THE aggressive downscaling of the transistor critical dimensions have been the way forward to keep the Moore's law valid [1], [2]. Moving from planar to 3-D architecture for better short-channel gate control, not only

increases the impact of process variations but also increases the number of available device parameters for performance optimization [3], [4]. Particularly for gate-all-around field-effect transistor (GAAFET), it is crucial to come up with an optimized trade-offs between the input design parameters (i.e., geometry, doping, and biases) and device outputs including speed and leakage. The understanding of optimum designs are necessary for improving the logic and RF performance in sub-2 nm technology node GAAFETs.

Some of the previous reported studies have focused on optimizing GAAFET performance by considering parameters such as channel material, geometry, and stress [4]. For instance, in case of an extremely scaled GAAFET, the susceptibility of carriers mobility on nanosheet (NS) width is studied [4]. In addition, quantum confinement (QC) effects are also examined in detail, which becomes dominant in ultra-thin NSs [5]. Thus, as we move from planar to GAA architecture, more complex nonlinear relationship of FET terminal characteristics on the device geometry and other design parameters have become evident.

Traditionally, TCAD has been used to accurately capture the complex nonlinear relationship between the FET parameters (geometry, doping, and bias) and the outputs of interest. The variability analysis is often performed on TCAD, and an optimized structure meeting target outputs are obtained [6]. This seems to work well for limited input parameter sets and simpler architectures. However, with GAAFET, the increase in input design parameters has resulted in a twofold problem.

- 1) It becomes time-consuming with the huge number of calculations to account for all physics in the input-output relationship.
- 2) In a multidimensional space, design exploration requires a large number of simulations at massive time costs. Even for an experienced user, the former is a hit and trial method without a proper framework.

To overcome the aforementioned problems, multiobjective optimization (MOO) design exploration of a GAAFET in a multidimensional space is a possible alternative. Previously, MOO has been used in various engineering fields for optimization of the multiobjective outputs [7], [8], [9], [10], [11], [12].

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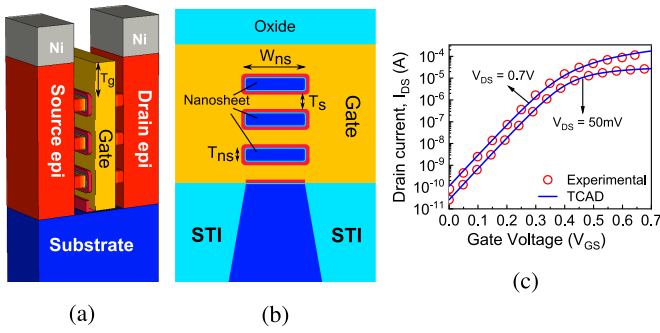


Fig. 1. (a) Three-dimensional structure and (b) cross-sectional cut of the GAAFET. (c) Calibration of the transfer characteristics of the GAAFET from a fabricated NS device [20].

Janet et al. [7] used the global optimization in transition metal complexes; Fu et al. [8] applied it in a two-stage transmission system. In particular, for the semiconductor process, it has been used for VLSI circuits [9], 2-D material FET [10], and Schottky-barrier power diode [11]. Xu et al. [12] used this for optimizing dc characteristics with channel geometry and doping profile as two input parameters of interest.

In this work, we have developed an automated MOO framework that address the design optimization challenges at an advanced technology node of a sub-2 nm GAAFET. The framework optimizes both the logic and RF performance metrics and consists of two parts: 1) a prior knowledge input difference artificial neural network (PKID ANN) surrogate model [13] mimicking the device and 2) the nondominated sorting genetic algorithm (NSGA-III) [14], and the strength Pareto evolutionary algorithm (SPEA2) [15], which generates the Pareto front and performs the final optimization. The PKID ANN captures the FET terminal characteristics as an analytical function of the device (geometry, doping, and bias) parameters [16], [17], [18], [19]. Further, it is used in lieu of repeated and costly TCAD simulations which helps multiobjective optimizers to perform fast and efficient design trade-offs.

This article is arranged in the following sections. In Section II, the development of TCAD setup, PKID ANN model, MOO framework, and algorithms are explained. In Section III, the results are discussed. Finally, Section IV concludes the work.

II. OPTIMIZATION METHODOLOGY

A. TCAD Simulation

A 3-D GAAFET structure is simulated on Sentaurus TCAD (U-2022.12-SP1), as shown in Fig. 1(a) and (b). The device characteristics are calibrated with the fabricated GAAFET having aggressively scaled L_g of 12 nm and contact poly-pitch (CPP) of 44 nm [20] of a sub-5 nm node. Gate-stack comprises of realistic SiO_2 and HfO_2 combination having EOT of 0.9 nm. Gate work function (GWF) is engineered to match the threshold voltage. The models used during the calibration include the vertical-field-dependent mobility model, high-field saturation mobility model, and gate-length-dependent ballistic mobility mode [21]. The ballistic mobility model default coefficients values for electrons and holes has

TABLE I
DEVICE PARAMETERS WITH A RANGE OF VARIATION

Symbol	Parameters	Range
N_{sd}	Source/Drain doping	$1.8 - 2.2e20/\text{cm}^3$
N_{ch}	Channel doping	$1.8 - 2.2e18/\text{cm}^3$
L_g	Gate Length	12 – 16nm
GWF	Gate work function	4.27 – 4.72eV
W_{ns}	GAA NS Width	24 – 36nm
T_{ns}	GAA NS Thickness	5 – 7nm
T_g	Top-gate Thickness	10 – 20nm
T_s	GAA NS Separation	5 – 10nm
V_{gs}	Gate-source voltage	0 – 0.65V
V_{ds}	Drain-source voltage	0 – 0.65V

been used. Shockley–Read–Hall (SRH) recombination model, and band-gap narrowing model with default parameters has also been used. Fermi statistics for carrier dynamics in highly doped regions is also used. Additionally, contact resistance model is used to fine-tune the terminal characteristics. Since NS thickness is small, we have used density gradient model to account for the QC effects. In addition, to emulate a realistic case, rounded corners of 1 nm for NSs are also used.

A sufficiently good match in $I-V$ characteristics is obtained between the experimental and simulated data, as shown in Fig. 1(c). The TCAD model acts as a reference GAAFET for dataset generation. After calibration, structural parameters of interest that significantly affects the logic/RF performance are identified and used as design parameters for the PKID ANN. Keeping in view the International Roadmap of Devices and Systems (IRDS) targets for 2025 [22], variation in various input parameters are kept within the range for a sub-2 nm node GAAFET and are shown in Table I.

B. GAAFET PKID ANN Model

The conventional ANN (c-ANN) is the most common form of feed-forward network; however, it requires a massive amount of training data which translates into a massive number of TCAD simulations. To mitigate this, a form of knowledge-based ANN (KBANN) known as the PKID ANN [13] has been used in this work [Fig. 2(a)]. Two distinct features of the model are: 1) a low-fidelity (LF) model to guide the training of ANN and 2) the target output is the error term $e(\lambda, y_{lf}(\lambda))$ (i.e., the difference between the TCAD and LF data). This is done to minimize the variance in the target output. The final output is obtained by adding the error term $e(\lambda, y_{lf}(\lambda))$ and the LF output $y_{lf}(\lambda)$.

A multilayer ANN is used, as shown in the zoomed out portion of Fig. 2(a). Each sample row has ten parameters that goes as input of both the ANN and LF model. The two target outputs are error in drain current error (I_d) and error in gate capacitance error (C_{gg}), respectively. The ANN has a single input layer of structural and bias parameters (a row vector with size 10×1) and a single output layer (a row vector with a size 2×1). Two hidden layers with a maximum of 30 neurons and a hyperbolic tangent activation function is used to accurately capture the nonlinearity between input-output. Data normalization is carried out for each parameter independently by computing the mean and the standard deviation. Once this is done, datasets are split into training, testing, and validation samples. The accuracy of PKID ANN is quantified

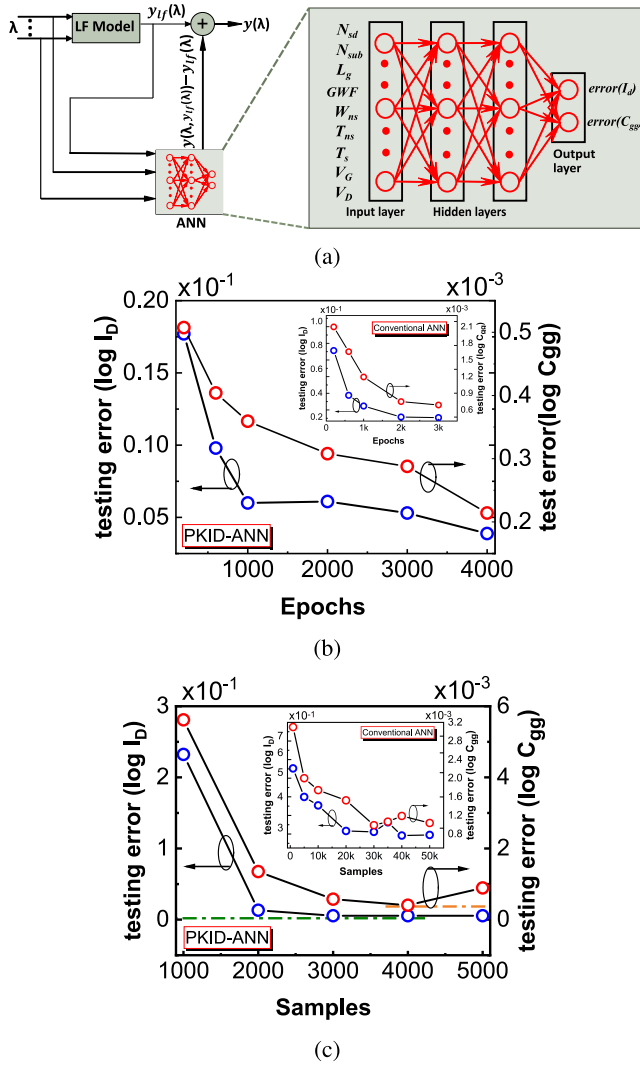


Fig. 2. (a) Architecture of the PKID ANN. Decay plot of rms testing error of $\log(I_d)$ and $\log(C_{gg})$ with (b) number of epochs and (c) number of samples. For comparison, the inset shows the same for c-ANN model. The minimum mean error threshold in case of c-ANN model is reached at much higher number of samples compared to PKID ANN, as evident in Fig. 2(c).

by the root mean square (rms) error for a fixed testing dataset of 5000 samples. For training the PKID ANN, progressive number of samples, $N_t = \{1000, 2000, 3000, 4000, 5000\}$, are used to study the optimal training criteria. The training samples points refers to the number of GAAFET devices. All the training points are uniformly distributed throughout the input parameter space of Table I using the Latin hypercube sampling (LHS). The rms of both train and test datasets kept decreasing as epochs and training samples grew, shown in the decay plot of Fig. 2(b) and (c), respectively. From the decay plots, assuming minimum mean error threshold of 0.2% for $I-V$ and 0.02% for the $C-V$ characteristics, the best possible hyperparameters for the best output accuracy are adequately chosen. As evident from Fig. 2(c), c-ANN fails to reach the target accuracy whereas PKID ANN meets the acceptable error benchmark at the same number of fixed samples. The number of samples, neurons, and epochs are 4000, 27, and 4000, respectively. The validation of the PKID ANN prediction with the true values of $I_{ds}-V_{gs}$, $I_{ds}-V_{ds}$, $C_{gg}-V_{gs}$, $C_{gg}-V_{ds}$, g_m-V_{gs} ,

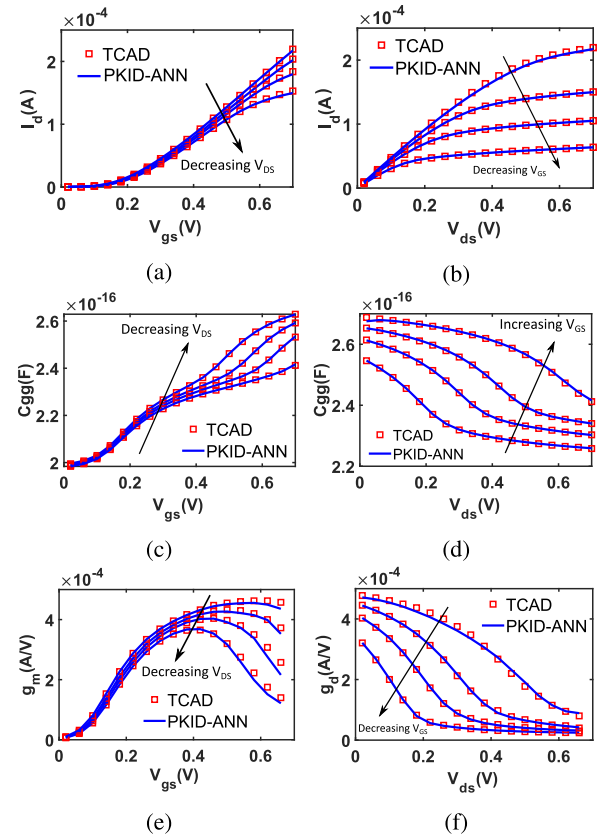


Fig. 3. Predicted ANN and true TCAD values for (a) I_d-V_g , (b) output characteristics, (c) $C_{gg}-V_g$ and (d) $C_{gg}-V_d$, and (e) g_m-V_g and (f) g_d-V_d . The terminal characteristics plotted are for the corner point $N_{sd} = 1.8 \times 10^{20} \text{ cm}^{-3}$, $N_{ch} = 1.8 \times 10^{18} \text{ cm}^{-3}$, $L_g = 12 \text{ nm}$, $GWF = 4.275 \text{ eV}$, $W_{ns} = 24 \text{ nm}$, $T_{ns} = 5 \text{ nm}$, $T_g = 10 \text{ nm}$, $T_s = 5 \text{ nm}$. For V_{gs} sweep, fixed V_{ds} values of 0.15, 0.35, 0.45, and 0.55 V are used, while for V_{ds} sweep, fixed V_{gs} values of 0.15, 0.35, 0.45, and 0.55 V are used.

and g_d-V_{ds} characteristics are shown in Fig. 3. Here, we have taken the corner points as our validation points. Corner points are the combination of extrema of all the input parameters at which the predicted outputs are tested. This is done to test the accuracy of the model at the far end of the input variation as an extreme case scenario. It is observed that the model predicts well and accurately the target outputs.

C. NS Design Objectives for MOO

GAAFETs are complex transistors with a high degree of nonlinear performance metrics. I_{ON}/I_{OFF} ratio and intrinsic delay (τ) are good metrics for logic operation. I_{ON}/I_{OFF} ratio is extracted from a trained ANN model that predicts the transfer characteristics. Likewise, intrinsic delay is calculated as $\tau = C_{gg}V_{DD}/I_{ON}$ [23]. Here V_{DD} is the supply voltage, I_{ON} and C_{gg} are, respectively, the ON-state current and gate capacitance. Moreover, the reciprocal of intrinsic delay ($1/\tau$) determines the speed of the GAAFET. To assess the RF performance, two additional metrics of intrinsic gain and cut-off frequency are chosen. Intrinsic gain is calculated as $g_m r_o$, where $r_o = ((\partial I_d / \partial V_{ds}))^{-1}$. This is a highest voltage gain which a transistor can provide when operated in saturation mode. Note that the frequency dependence of C_{gg} , g_m , and gain are not modeled in the present PKID ANN for simplicity. This limits the maximum operating frequency up to 50 GHz ahead

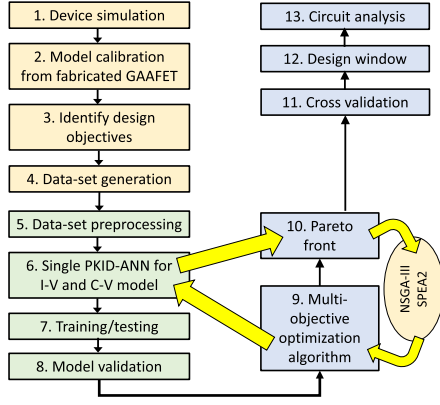


Fig. 4. MOO design framework for an advanced GAAFET design parameters.

TABLE II
SUB-2 NM NODE GAAFET PARAMETERS

Parameters	IRDS target (2025)	Variation in this work
Gate Length	14 nm	12-16 nm
Gate Pitch	45 nm	43-47 nm
Device effective width	216 nm	174-258 nm
Lateral GAA vertical pitch	18 nm	14.6-21.6 nm
Lateral GAA NS thickness	6 nm	5-7 nm
Gate height over fin	15 nm	10-20 nm

of which the parameters will become frequency dependent [6]. Here, both g_m and r_o are obtained from the derivative of $I_{ds}-V_{gs}$ and $I_{ds}-V_{ds}$ coming from the PKID ANN operating in saturation mode. As evident from the Fig. 3(d), maximum g_m may not necessarily occur at fixed bias. Thus, we have calculated the gate bias at which the maximum g_m occurs, and simultaneously calculated the smallest value of $g_d = 1/r_o$ at same V_{gs} , and highest V_{ds} . Cut-off frequency, on the other hand, is the frequency at which the small-signal current gain becomes unity. It is given as $f_T = g_m/2\pi C_{gg}$, where maximum g_m and minimum C_{gg} are extracted from the PKID ANN. The upper and lower bounds of the various parameters have been chosen in accordance with the IRDS target for 2025 [22], as shown in Table II.

D. MOO Framework and Algorithms

Fig. 4 describes the overall MOO design flow for GAAFET. It consists of several parts: 1) TCAD simulation and calibration. Identification of the input parameters and multiple target objectives; 2) dataset generation, model development; 3) spontaneous prediction of the $I - V$ and $C - V$ within the desired range; 4) MOO to obtain the Pareto front using genetic algorithms (GAs); 5) iteratively return to step 3) till the convergence criteria meets; 6) design window from the Pareto front; 7) cross validation of Pareto front with TCAD; and 8) finally, the circuit-level analysis.

Two different GAs NSGA-III [14], [24] and SPEA2 [15], [25] have been used. The working of the GAs are mentioned and compared in Table III. The algorithms aim to find a set of nondominated solutions (i.e., target objectives). The new population is calculated iteratively until the accuracy threshold is reached or maximum iteration criteria are met. Both NSGA-III

TABLE III
GAs USED IN MOO

Algorithm1: NSGA-III [14]	Algorithm2: SPEA2 [15]
1. Set $k=0$ 2. Initialize P_0 and Q_0 3. while $k < G$ do a. Calculate fitness for P_k and assign rank based on Pareto dominance b. Perform selection on P_k to fill mating pool c. Apply crossover and mutation operators to obtain the offspring Q_k d. Select the best N non-dominated solutions from $P_k \cup Q_k$ by the two-step procedure to form P_{k+1} e. Set $k = k + 1$ 4. end 5. return P_k	1. Set $k=0$ 2. Initialize P_0 and A_0 3. while $k < G$ do a. Calculate fitness for P_k and A_k b. Copy all non-dominated solutions from $P_k \cup A_k$ to A_{k+1} c. Use the truncation operator to remove elements from A_{k+1} d. Use dominated individuals from A_k to fill A_{k+1} , if the capacity of A_{k+1} has not exceeded. e. Perform selection on A_{k+1} to fill mating pool f. Apply recombination and mutation operators to obtain P_{k+1} g. Set $k = k + 1$ 4. end 5. return A_k

and SPEA2 use identical population and archive size of 30. The crossover percentage P_{cross} is selected from $\{0.3, 0.5, 0.7\}$ and mutation percentage P_{mut} varies in $\{0.3, 0.5\}$. Multiple combination of these parameters are run to record the final population. In all cases, we observed better Pareto-optimal solution and diversity in SPEA2 compared to the NSGA-III. This is possibly due to the better elitism prevention strategy in SPEA2 [26].

III. RESULTS AND DISCUSSION

A. MOO for GAAFET Logic Parameters

Two design objectives, I_{ON}/I_{OFF} ratio and intrinsic speed ($1/\tau$) for a better logic switch have been optimized in this section. The automatic MOO design flow has been used to obtain the Pareto front that maximizes both the objectives. We observed that increasing the I_{ON}/I_{OFF} ratio decreases the speed of the transistor, as evident from Fig. 5(a). Since the SPEA2 algorithm gives a better Pareto front, both in terms of diverse solutions as well as it being closer to the target, we focus on the distribution obtained from it. The distributions of the input design parameters in the Pareto front are shown in Fig. 5(b) and (c). It can be identified that smaller L_g (≈ 12 nm) is preferred for both NSGA-III and SPEA2. GWF covers mid-to-higher ranges for both algorithms as it depletes the channel of excess carriers in the OFF-state and increases the threshold voltage of the GAAFET. From a closer examination of the Pareto front, a higher I_{ON}/I_{OFF} ratio of $\geq 10^7$ requires GWF to be higher than 4.6 eV. T_g and T_s are important parameters that determine the gate capacitance and have a lower mid-value distribution for SPEA2. For the NS width and thickness, it can be seen that they both are on the higher side since I_{ON} is proportional to them. Both design objectives require a higher I_{ON} for a better I_{ON}/I_{OFF} ratio and speed. The results in Fig. 5 provides the optimum design trade-offs between them.

B. MOO for GAAFET RF Parameters

The RF performance of the GAAFET is next investigated using the MOO design flow. Fig. 6(a) shows the obtained Pareto front between the two design objectives, i.e., gain ($g_m r_o$) and cut-off frequency ($g_m/2\pi C_{gg}$). Both objectives

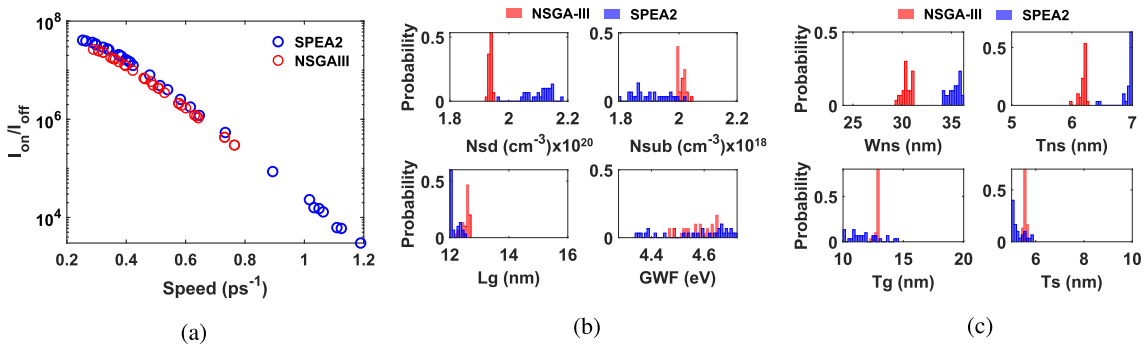


Fig. 5. (a) Comparison of Pareto front obtained while maximizing the I_{ON}/I_{OFF} and intrinsic speed ($1/\tau$) as two objectives with NSGA-III and SPEA2 algorithms. (b) and (c) Comparison of the input design parameters distribution in the Pareto front.

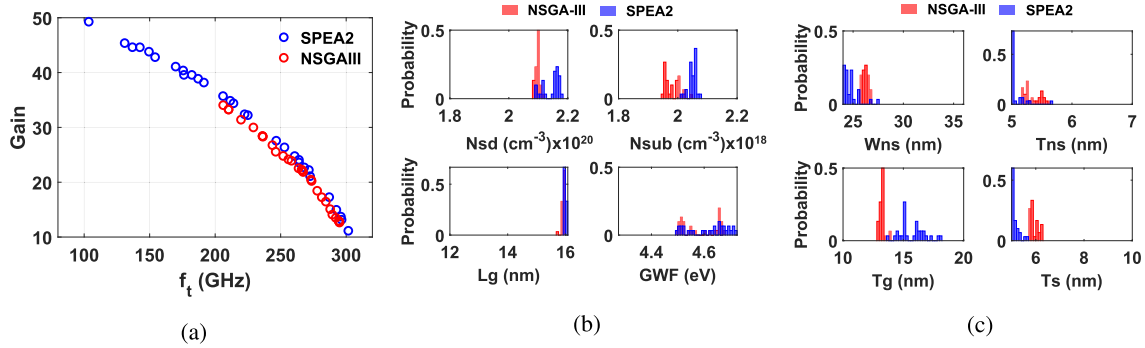


Fig. 6. (a) Comparison of Pareto front obtained while maximizing the gain and cut-off frequency (f_T) as two objectives with NSGA-III and SPEA2 algorithms. (b) and (c) Comparison of the input design parameters distribution in the Pareto front.

should have a higher value meaning the target space is located at the top right corner. Again, the SPEA2 offers a better Pareto front than the NSGA-III, whose distribution is limited. Unlike logic operation that requires the lower bound of the L_g , here, higher bound of L_g is favored for both algorithms, as evident from Fig. 6(b). In doping, a higher bound of N_{sd} and mid-range of N_{ch} is identified for both SPEA2 and NSGA-III. GWF variation is similar to the previous case; however, with closer examination, we observe that mid-range GWF corresponds to the higher (≥ 290 GHz) in the Pareto front while larger GWF corresponds to the higher gain (≥ 40). Both W_{ns} and T_{ns} is around the smaller bound [Fig. 6(c)]. Aside from this, top gate thickness varies in the mid-range for all the values of target objectives from the Pareto front. From the above analysis, we find certain design parameters (L_g , W_{ns} , etc.) distributed at extreme ends of the SPEA2 Pareto front for both the logic and RF. This MOO flow, thus, provide us the insight for designing sub-2 nm node GAAFET depending upon the application in perspective.

C. Pareto Front Cross-Validation With TCAD

To validate the input parameters obtained from the SPEA2 Pareto front, design objectives calculated at those points from TCAD are shown in Fig. 7(a) and (b), respectively. We observe similar trends for the design objectives from both the SPEA2 Pareto front and TCAD simulations. In fact, both the predicted and true value of I_{ON}/I_{OFF} and speed are in close agreement, whereas for f_T and gain, the deviation is marginal which can be attributed to a slightly different value predicted by the PKID ANN for derivative parameters (g_m and r_o) compared to the

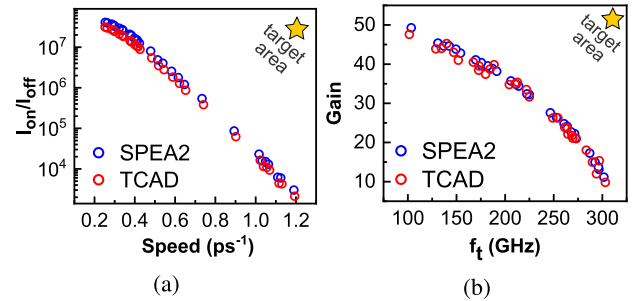


Fig. 7. Design objectives from SPEA2 and TCAD simulations for cross-validation. (a) I_{ON}/I_{OFF} versus speed and (b) gain versus f_T . PKID ANN model successfully acts as a surrogate for TCAD.

TCAD. This is also evident from the corner point validation results in Fig. 3.

D. GAAFET Optimum Design Window

The desired input parameters for optimized logic/RF parameters are shown in Figs. 5 and 6, respectively. From the Pareto front solutions, one can extract the optimum design window for the input parameters. The aim is to get closer to the I_{ON}/I_{OFF} ratio and f_T from IRDS projection for “2 nm” GAAFET (2025). Table IV lists the various optimized parameters. We infer that a projected I_{ON}/I_{OFF} of $\sim 10^5$ requires L_g to be on the smaller end for better logic. Likewise, the target values of speed is also in close agreement with the IRDS projection. Simultaneously, target f_T requires L_g to be on higher end for better RF performance. Finally, it can be said that by careful design of device geometrical parameters, its logic/RF performance can be improved to meet multiple IRDS targets for future technology nodes.

TABLE IV
OPTIMIZED GAAFET PARAMETERS (IRDS 2025
PROJECTION VERSUS SPEA2 AND TCAD)

	Parameters	2025 IRDS	Logic	RF
Target Objectives	I_{on}/I_{off}	$\sim 10^5$	SPEA2 8.9×10^4 TCAD 9×10^4	-
	$1/\tau$ (ps $^{-1}$)	1.04	SPEA2-0.9 TCAD-0.9	-
	Gain (dB)	40	-	SPEA2-30.1 TCAD-30
	f_T (GHz)	304	-	SPEA2-224.9 TCAD-220.4
Design Parameters	L_g	14 nm	12 nm	16 nm
	GW_F	-	4.45 eV	4.55 eV
	W_{ns}	30 nm	35 nm	24 nm
	T_{ns}	6 nm	7 nm	5 nm
	T_g	15 nm	11 nm	16.5 nm
	T_s	7.4 nm	5 nm	5 nm

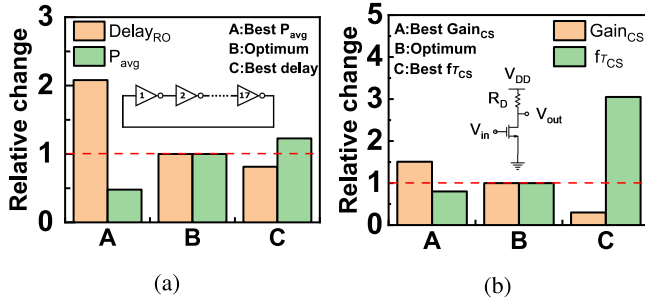


Fig. 8. (a) Relative change in the delay per stage (delay_{RO}) and average power consumption (P_{avg}) of a 17-stage ring oscillator for best cases and optimized GAAFET. (b) Relative change in the gain $_{CS}$ and f_{TCS} of a common source amplifier for best cases and optimized GAAFET.

E. Circuit Analysis

We further verified the accuracy of the optimized parameters from SPEA2 pareto front via circuit simulations. For this purpose, the developed PKID-ANN model has been converted into a SPICE compatible Verilog-A model, which is then used to run circuit simulations. For the case of I_{ON}/I_{OFF} and speed ($1/\tau$), a 17-stage ring oscillator is designed for three different GAAFET (A, B, C) geometries. An improved I_{ON}/I_{OFF} implies low P_{avg} dissipation while a smaller delay reflects better speed. The three geometries are chosen from Pareto front [Fig. 7(a)] such that structure-A gives the best P_{avg} , structure-B is the optimized one [Table IV], and structure-C gives the best delay. The relative change in the targets compared to the optimized structure-B is evident from the Fig. 8(a). Similarly, the relative change in gain $_{CS}$ and f_{TCS} obtained from the common-source amplifier is shown in the Fig. 8(b). Again, the three geometries are chosen from the Pareto front [Fig. 7(b)]. The geometry with optimized parameters is obtained for the structure-B. We observed one-to-one mapping in the input-output relation both at the device and circuit-level analysis.

Thus, the circuit simulation further consolidates that the optimized logic/RF parameters can be obtained from the input design parameters of the SPEA2 Pareto front.

F. Benefits of Proposed Framework

It is worth mentioning that our PKID ANN-based MOO framework offers several advantages compared to the TCAD-based traditional optimization routines.

- 1) Once PKID ANN is trained, the time-cost savings of repeated simulations are huge for optimizing different target objectives.
- 2) The MOO framework is perfectly compatible with any ANN architecture [13], [18], [19], [27], [28] and the developers/users are free to choose an ANN that suits their accuracy/speed/computational efficiency.
- 3) Integrating TCAD in MOO is difficult since GAs require analytical models and thus, automation is difficult. The proposed framework, on the other hand allows easy automation for devices/targets of interest, resulting in significant cost savings.
- 4) PKID ANN has the advantage of being Verilog-A compatible and can be combined with interconnect SPICE models to perform circuit-level MOO optimizations including multiple device and interconnect effects.

IV. CONCLUSION

In this work, an machine learning (ML)-augmented MOO design flow has been developed for design optimization of advanced node transistors. The proposed framework utilizes analytical PKID ANN model in conjunction with GAs. This has multiple advantages compared to conventional TCAD-based optimization routines such as: 1) significant reduction of computational cost of repeated simulations for optimizing different target objectives; 2) compatibility with genetic optimization algorithms allowing easy automation and plug-and-play facility for devices/targets of interest; 3) compatibility with any ANN architecture making it future-proof, and finally; and 4) SPICE compatible implementation allowing the proposed framework to be used for circuit-level performance optimizations. The detailed working and validation of the proposed flow has been successfully demonstrated for a stacked NS GAAFET structure, where multiple target design optimizations objectives for both logic and RF were considered. We have shown that our MOO framework allows us to optimize the device such that multiple logic and RF target objectives can meet the IRDS projection for future technology nodes. This framework has the potential to speed-up and automate device design and optimization for modern FETs and circuits leading to significant benefits for the community.

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